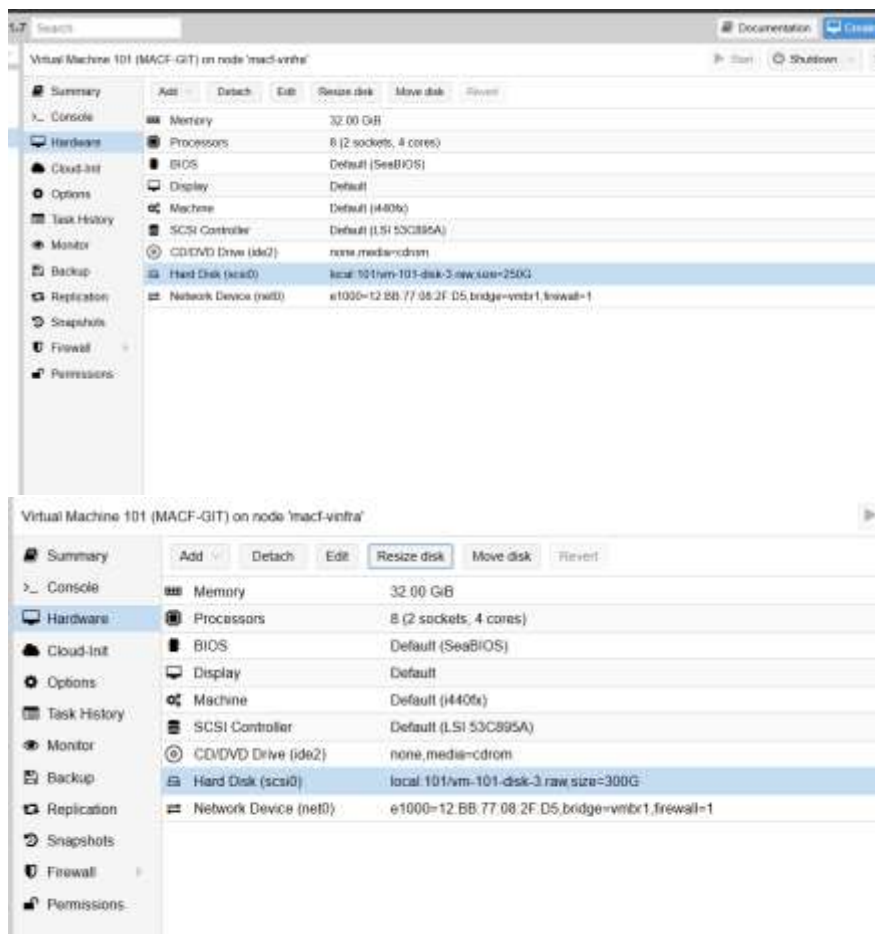


Extend Disk Linux from /Data

```
root@git:/home/usradmsvr# df -kh
Filesystem      Size  Used Avail Use% Mounted on
udev            16G   0    16G   0% /dev
tmpfs           3.2G  17M   3.2G   1% /run
/dev/sda2       98G   88G   5.1G  95% /
tmpfs           16G   28K   16G   1% /dev/shm
tmpfs           5.0M   0    5.0M   0% /run/lock
tmpfs           16G   0    16G   0% /sys/fs/cgroup
/dev/loop0      56M   56M   0 100% /snap/core18/2887
/dev/loop2      64M   64M   0 100% /snap/core20/2599
/dev/loop3      64M   64M   0 100% /snap/core20/2582
/dev/loop1      56M   56M   0 100% /snap/core18/2855
/dev/loop7      74M   74M   0 100% /snap/core22/2010
/dev/loop4      74M   74M   0 100% /snap/core22/1981
/dev/loop5      67M   67M   0 100% /snap/core24/1006
/dev/loop6      67M   67M   0 100% /snap/core24/988
/dev/loop8      112M  112M   0 100% /snap/lxd/33246
/dev/mapper/gitlab-data 149G  119G   31G  80% /data
/dev/loop9      114M  114M   0 100% /snap/lxd/34099
/dev/loop10     51M   51M   0 100% /snap/snapd/24505
/dev/loop12     51M   51M   0 100% /snap/snapd/24718
tmpfs           3.2G   0    3.2G   0% /run/user/1000
root@git:/home/usradmsvr#
```

1. Resize disk from 250 to 300



2. Rescan disk and check the partition

```
Get DMU size: dmsetup (524285952 -> 629145600) will be corrected by write.
The backup GPT table is not at the end of the device: this problem will be corrected by write.
Disk /dev/sda: 300 GiB, 322122547200 bytes, 629145600 sectors
Disk model: QEMU HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 4781D841-D67C-4C53-894B-3BC79D80BBFD

Device      Start      End      Sectors  Size Type
/dev/sda1    2048      4095      2048     1M BIOS boot
/dev/sda2    4096 209713151 209709056 100G Linux filesystem
/dev/sda3   209713152 524285951 314572800 150G Linux filesystem

Disk /dev/mapper/gitlab-data: 149 GiB, 159987531776 bytes, 312475648 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

3. Add new partition

```
Command (m for help): n
Partition number (4-128, default 4):
First sector (524285952-629145566, default 524285952):
Last sector, +/-sectors or +/-size[K,M,G,T,P] (524285952-629145566, default 529145566): +50G

Created a new partition 4 of type 'Linux filesystem' and of size 50 GiB.

Command (m for help): p
Disk /dev/sda: 300 GiB, 322122547200 bytes, 629145600 sectors
Disk model: QEMU HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 4781D841-D67C-4C53-894B-3BC79D80BBFD

Device      Start      End      Sectors  Size Type
/dev/sda1    2048      4095      2048     1M BIOS boot
/dev/sda2    4096 209713151 209709056 100G Linux filesystem
/dev/sda3   209713152 524285951 314572800 150G Linux filesystem
/dev/sda4   524285952 629143551 104857600  50G Linux filesystem

Command (m for help):
```

4. Add Physical Volume

```
root@git:/home/usradmsvr#
root@git:/home/usradmsvr# pvcreate /dev/sda
sda sda1 sda2 sda3 sda4
root@git:/home/usradmsvr# pvcreate /dev/sda4
Physical volume "/dev/sda4" successfully created.
root@git:/home/usradmsvr#
root@git:/home/usradmsvr#
root@git:/home/usradmsvr#
```

5. Extend VG from new PV

```
root@git:/home/usradmsvr#
root@git:/home/usradmsvr# vgextend gitlab /dev/sda4
Volume group "gitlab" successfully extended
root@git:/home/usradmsvr#
```

6. Add 20G to LVM from VG

```
data gitlab -wi-ao---- 149.00g
root@git:/home/usradmsvr# vgs
VG      #PV #LV #SN Attr   VSize  VFree
gitlab  2   1  0 wz--n- 199.99g 50.99g
root@git:/home/usradmsvr# lvs
LV VG      Attr   LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
data gitlab -wi-ao---- 149.00g
root@git:/home/usradmsvr#
root@git:/home/usradmsvr#
root@git:/home/usradmsvr#
root@git:/home/usradmsvr# lvextend -L +20G /dev/gitlab/data
Size of logical volume gitlab/data changed from 149.00 GiB (38144 extents) to 169.00 GiB (43264 extents).
Logical volume gitlab/data successfully resized.
root@git:/home/usradmsvr#
```

7. Rescan mount point depends file system (xfs or ext4)

```

root@git:/home/usradmsvr# xfs_growfs /dev/mapper/gitlab-data
meta-data=/dev/mapper/gitlab-data isize=512    agcount=4, agsize=9764864 blks
        =                       sectsz=512    attr=2, projid32bit=1
        =                       crc=1        finobt=1, sparse=1, rmapbt=0
        =                       reflink=1
data      =                       bsize=4096   blocks=39059456, imaxpct=25
        =                       sunit=0       swidth=0 blks
naming    =version 2             bsize=4096   ascii-ci=0, ftype=1
log       =internal log         bsize=4096   blocks=19072, version=2
        =                       sectsz=512    sunit=0 blks, lazy-count=1
realtime  =none                 extsz=4096   blocks=0, rtextents=0
data blocks changed from 39059456 to 44302336
root@git:/home/usradmsvr#
root@git:/home/usradmsvr#
root@git:/home/usradmsvr#
root@git:/home/usradmsvr# df -kh
Filesystem      Size  Used Avail Use% Mounted on
udev             16G   0    16G   0% /dev
tmpfs            3.2G  17M   3.2G   1% /run
/dev/sda2        98G   88G   5.1G  95% /
tmpfs            16G   28K   16G   1% /dev/shm
tmpfs            5.0M   0    5.0M   0% /run/lock
tmpfs            16G   0    16G   0% /sys/fs/cgroup
/dev/loop0       56M   56M   0 100% /snap/core18/2887
/dev/loop2       64M   64M   0 100% /snap/core20/2599
/dev/loop3       64M   64M   0 100% /snap/core20/2582
/dev/loop1       56M   56M   0 100% /snap/core18/2855
/dev/loop7       74M   74M   0 100% /snap/core22/2010
/dev/loop4       74M   74M   0 100% /snap/core22/1981
/dev/loop5       67M   67M   0 100% /snap/core24/1006
/dev/loop6       67M   67M   0 100% /snap/core24/988
/dev/loop8       112M  112M   0 100% /snap/lxd/33246
/dev/mapper/gitlab-data 169G  119G   51G   71% /data
/dev/loop9       114M  114M   0 100% /snap/lxd/34099
/dev/loop10      51M   51M   0 100% /snap/snapd/24505
/dev/loop12      51M   51M   0 100% /snap/snapd/24718
tmpfs            3.2G   0    3.2G   0% /run/user/1000
root@git:/home/usradmsvr#

```